

Pocket 3 PTZ Control (USB-C / UVC)

English | [繁體中文](#)

Turn a DJI Osmo Pocket 3 into a remote-controllable PTZ camera over a single USB-C cable. It uses Windows DirectShow UVC camera control (the same path behind OBS's "Camera Control" sliders) to drive pan / tilt / zoom in real time — **without taking over the video stream**, so you can output the camera in OBS and control the gimbal at the same time. Bluetooth (BLE/DUML) does **not** work for gimbal control on the Pocket 3, which is why this project uses the UVC route.

1. Setup (once per machine)

1. **Install Python 3.12 (64-bit)** from <https://www.python.org/downloads/> and tick "Add Python to PATH". (The OBS native-script option only accepts 3.12.)
2. **Install packages:**

```
pip install pygrabber comtypes keyboard
```

3. **Connect the camera:** plug the Pocket 3 in via USB-C and choose **Webcam mode** on the camera screen.

2. Four ways to use it

A. OBS dock panel (recommended)

1. Double-click `start-ptz.bat`. A console shows "PTZ panel started" and auto-connects. **Keep that window open** (it is the control core).
2. In OBS: **Docks** → **Custom Browser Docks**, add one with URL:

```
http://127.0.0.1:8723/
```

3. A panel appears inside OBS with an analog joystick, Center, Zoom, and Sensitivity.

B. Standalone GUI

Double-click `start-gui.bat`, or:

```
python uvc-gui.py
```

Analog joystick, live values, customizable keys (click a key button, then press a key), and a **language switch (中文 / English)** in the top-right.

Shared settings: the GUI and the OBS panel share `ptz-config.json`, so sensitivity, center angle, and the chosen camera carry over. Custom keys are GUI-only (the panel uses buttons).

C. Numpad control

```
python uvc-numpad.py
```

Numpad: `[8/2/4/6]` = tilt/pan, `[5]` = center, `[7/9]` = zoom, `[+/-]` = speed, `[/]` = quit.

D. OBS native hotkeys

Load `pocket-ptz-obs.py` via OBS **Tools** → **Scripts** (first point Python Settings at your Python 3.12 folder; the bottom line must read `Python 3.12.x`). Assign keys under **Settings** → **Hotkeys**, search "Pocket".

3. Troubleshooting

- **Camera not auto-detected / wrong camera:** every tool lets you pick manually — the OBS panel has a camera dropdown +  rescan button at the top; the GUI has a device dropdown; `uvc-numpad.py` / `uvc-smooth.py` list all devices and ask you to type an index (or pass one: `python uvc-numpad.py 2`); the OBS script has a device list in its properties. Your choice is saved to `ptz-config.json`.
- **No camera / empty list:** confirm Webcam mode; confirm `pygrabber comtypes` are installed in the Python you actually use.
- **Connected but nothing moves:** run `python uvc-ptz2.py` to probe supported pan/tilt/zoom ranges.
- **OBS script shows "No properties available":** Python Settings still says "not loaded" — point it at the folder containing `python.exe`; it must show `Python 3.12.x`.
- **Center is off:** UVC pan unit is arc-seconds ($1^\circ = 3600$). Adjust `home_deg` (default $-90 = 90^\circ$ left of center).

4. Limitations

- **ActiveTrack cannot be triggered over USB** (a DJI restriction). Tap the subject on the camera screen.
- **Zoom is digital**, same as the camera itself.

5. Publish to GitHub

This folder already includes `.gitignore` (ignores each machine's `ptz-config.json`) and `LICENSE`.

Edit `LICENSE` and replace `[your name or GitHub handle]` first.

1. On github.com → **New repository** (e.g. `pocket3-usb-ptz`), do **not** add a README, click Create.
2. In this folder, run (swap in your URL):

```
git init
git add .
git commit -m "Pocket 3 USB-C PTZ control"
git branch -M main
git remote add origin https://github.com/<you>/pocket3-usb-ptz.git
git push -u origin main
```

3. Later changes: `git add . && git commit -m "..." && git push`

Files

File	Purpose
<code>ptz-server.py</code>	Background control program + web panel (for the OBS browser dock)
<code>start-ptz.bat</code>	Double-click launcher for <code>ptz-server.py</code>
<code>uvc-gui.py</code>	Standalone GUI (joystick, custom keys, language switch)
<code>start-gui.bat</code>	Double-click launcher for <code>uvc-gui.py</code>
<code>uvc-numpad.py</code>	Numpad control
<code>uvc-smooth.py</code>	Arrow-key control (smoothed)
<code>uvc-ptz2.py</code>	Probe tool: lists supported controls and ranges
<code>pocket-ptz-obs.py</code>	OBS native-hotkey script
<code>COMPANION.md</code>	Bitfocus Companion / Stream Deck guide

[](#)

Pocket 3 PTZ 控制 (USB-C / UVC)

[English](#) | 繁體中文

把 DJI Osmo Pocket 3 透過一條 USB-C 線變成可遙控的 PTZ 攝影機。走的是 Windows DirectShow 的 UVC 相機控制 (也就是 OBS 「相機控制」滑桿背後那條路) , 即時控制 pan /

tilt/zoom, **而且不佔用影像串流**——可以一邊用 OBS 出畫面、一邊遙控雲台。藍牙 (BLE/DUML) 在 Pocket 3 上實測無法控制雲台, 因此本專案走 UVC 這條可行路線。

一、安裝 (每台機器一次)

1. 裝 Python 3.12 (64-bit) , 到 <https://www.python.org/downloads/> 下載, 安裝時勾「Add Python to PATH」。(OBS 原生腳本那個選項只吃 3.12。)
2. 裝套件:

```
pip install pygrabber comtypes keyboard
```

3. 接相機: Pocket 3 用 USB-C 接電腦, 相機螢幕點進 **Webcam 模式**。

二、四種用法

A. OBS 控制面板 (最推薦)

1. 雙擊 `start-ptz.bat`, 黑視窗顯示「PTZ 控制面板已啟動」並自動連線。**這個視窗要一直開著** (它是控制核心)。
2. OBS: 「停駐視窗 → 自訂瀏覽器停駐視窗」, 新增一個, 網址:

```
http://127.0.0.1:8723/
```

3. OBS 裡會出現含類比搖桿、回正、Zoom、靈敏度的面板。

B. 獨立 GUI

雙擊 `start-gui.bat`, 或:

```
python uvc-gui.py
```

類比搖桿、即時數值、可自訂按鍵 (按按鍵後再按下要綁定的鍵), 右上有**語言切換 (中文 / English)**。

設定共用: GUI 與 OBS 面板共用 `ptz-config.json`, 靈敏度、回正角度、選的相機會沿用。自訂按鍵為 GUI 專屬 (面板用畫面按鈕)。

C. 小鍵盤控制

```
python uvc-numpad.py
```

小鍵盤: `8/2/4/6`=tilt/pan、`5`=回正、`7/9`=zoom、`+/-`=速度、`/`=離開。

D. OBS 原生快捷鍵

把 `pocket-ptz-obs.py` 用 OBS「工具 → 指令碼」載入（先在 Python 設定頁指到 Python 3.12 資料夾，下面要顯示 `Python 3.12.x`）。到「設定 → 快捷鍵」搜尋「Pocket」指定按鍵。

三、疑難排解

- **沒自動抓到相機 / 抓錯台：** 每個工具都能手動選——OBS 面板頂端有相機下拉選單 +  重新偵測；GUI 有裝置下拉選單；`uvc-numpad.py` / `uvc-smooth.py` 會列出所有裝置請你輸入編號（也可直接帶入：`python uvc-numpad.py 2`）；OBS 腳本的屬性裡也有裝置清單。選擇會存進 `ptz-config.json`。
- **找不到相機 / 清單空的：** 確認在 Webcam 模式；確認 `pygrabber comtypes` 裝在你實際用的那個 Python。
- **連到了但按了不動：** 跑 `python uvc-ptz2.py` 看 pan/tilt/zoom 支援與範圍。
- **OBS 腳本顯示「無可用的屬性」：** Python 設定頁還是「未載入」——要指到含 `python.exe` 的資料夾，看到 `Python 3.12.x` 才對。
- **回正不準：** UVC pan 單位是角秒（ $1^\circ=3600$ ）。調 `home_deg`（預設 -90 = 中心往左 90° ）。

四、限制

- **ActiveTrack 無法透過 USB 觸發**（DJI 限制），請在相機螢幕點主體。
- **變焦為數位變焦**，與相機本身一致。

五、上傳到 GitHub

本資料夾已含 `.gitignore`（忽略各機器自己的 `ptz-config.json`）與 `LICENSE`。

▮ 先把 `LICENSE` 裡的 `[your name or GitHub handle]` 改成你的名字。

1. github.com → **New repository**（例如 `pocket3-usb-ptz`），**不要勾** Add README，按 Create。
2. 在這個資料夾執行（網址換成你的）：

```
git init
git add .
git commit -m "Pocket 3 USB-C PTZ 控制工具"
git branch -M main
git remote add origin https://github.com/你的帳號/pocket3-usb-ptz.git
git push -u origin main
```

3. 之後改動：`git add . && git commit -m "說明" && git push`

檔案	用途
<code>ptz-server.py</code>	背景控制程式 + 網頁面板（給 OBS 自訂瀏覽器停駐視窗）
<code>start-ptz.bat</code>	雙擊啟動 <code>ptz-server.py</code>
<code>uvc-gui.py</code>	獨立 GUI（搖桿、可自訂按鍵、語言切換）
<code>start-gui.bat</code>	雙擊啟動 <code>uvc-gui.py</code>
<code>uvc-numpad.py</code>	小鍵盤控制
<code>uvc-smooth.py</code>	方向鍵控制（平滑版）
<code>uvc-ptz2.py</code>	探測工具：列出支援的控制項與範圍
<code>pocket-ptz-obs.py</code>	OBS 原生快捷鍵腳本
<code>COMPANION.md</code>	Bitfocus Companion / Stream Deck 指南